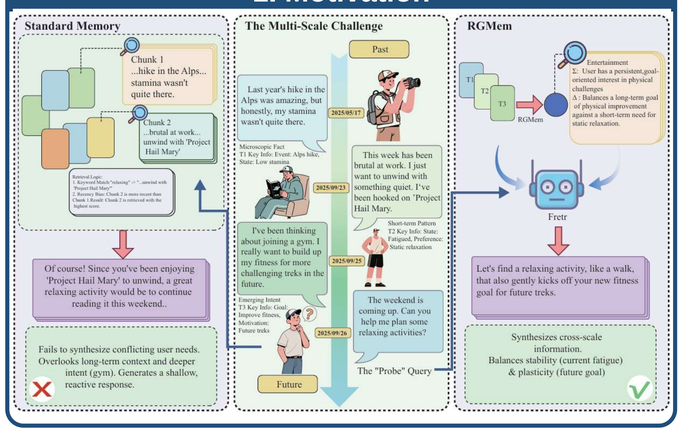
RGMem: Renormalization Group–inspired Memory Evolution for Language Agents

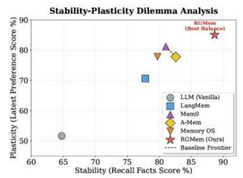
Ao Tian[1], Yunfeng Lu[1], Xinxin Fan[2] ,Changhao Wang[1], Lanzhi Zhou[1], Yeyao Zhang[3], Yanfang Liu[1]
[1]BeiHang University [2]Institute of Computing Technology [3]LightSail

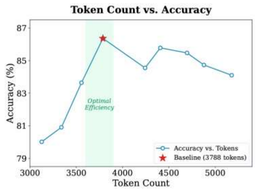
1. Motivation



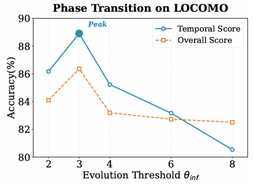
2. Insights

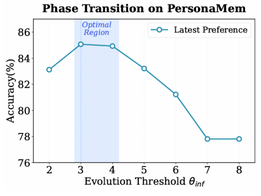
Insight 1. Effective Information Density is Maximized via Hierarchical Coarse-Graining.





Insight 2. User Profile Updates Exhibit Phase-Transition Like Dynamics.

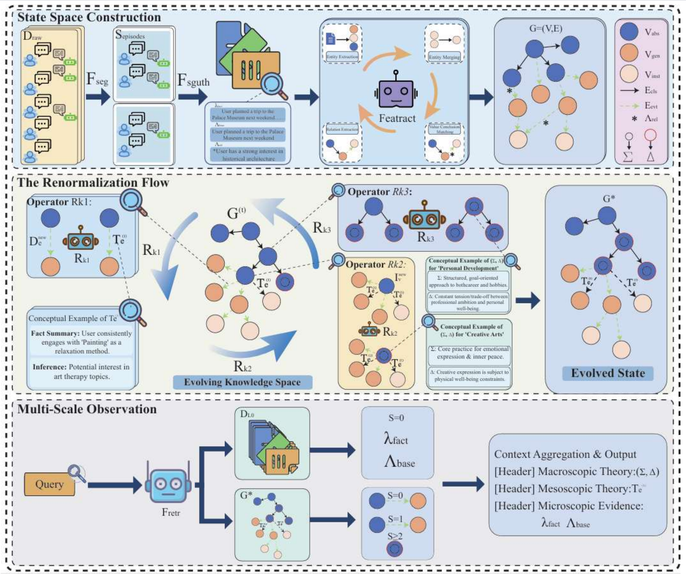




Insight 3. Separating Slow and Fast Variables Resolves the Stability–Plasticity Dilemma.

Insight 4. Long-Term User Profiles Exhibit Macroscopic Invariance Beyond Fact-Level Stability.

3. Method: RGMem



4. Main Results

Backbone	Method	Memory Recall: Recall Facts	Memory Recall: Suggest Ideas	Memory Recall: Revisit Reasons	Reasoning & Adaptation: Track Evol.	Reasoning & Adaptation: Aligned Rec.	Reasoning & Adaptation: General. Scen.	Temporal: Latest Pref.	Avg.
Backbone: GPT-4o-mini									
	LLM (Vanilla)	55.34	12.33	70.02	58.21	42.33	33.05	34.31	39.21
	LangMem	64.76	23.93	81.82	53.30	45.45	42.26	58.12	52.36
	Mem0	69.57	19.53	78.75	56.72	61.82	52.19	68.25	56.79
	A-Mem	59.78	9.22	80.19	53.30	44.32	45.42	53.25	49.17
	Memory OS	72.59	22.62	84.42	**69.11**	67.82	35.72	64.71	54.23
	RGMem	**77.06** (+4.47)	**26.47** (+2.54)	**85.29** (+0.87)	67.82	**73.66** (+5.84)	**56.62** (+4.43)	**75.47** (+7.22)	**63.87** (+7.08)
Backbone: GPT-4.1									
	LLM (Vanilla)	64.76	19.53	81.82	67.21	53.14	53.91	51.62	51.86
	LangMem	77.82	24.27	82.16	54.33	42.33	63.22	70.59	58.23
	Mem0	81.02	16.28	81.82	54.33	54.35	65.13	81.22	60.44
	A-Mem	82.57	28.31	86.35	56.72	70.12	65.13	77.81	63.95
	Memory OS	79.72	19.33	82.16	60.34	73.66	**74.18**	77.81	65.03
	RGMem	**88.64** (+6.07)	**35.04** (+6.73)	**87.03** (+0.68)	**70.89** (+3.68)	**83.02** (+9.36)	72.55	**85.07** (+3.85)	**74.01** (+8.98)

Method	Question Type: S-Hop	M-Hop	Open	Temp.	Avg.
gpt-4o-mini					
RAG	35.05	30.31	43.52	27.58	38.10
LangMem	62.23	47.92	71.12	23.43	58.10
Mem0	67.13	51.15	**72.93**	55.51	66.88
Zep	74.11	66.04	67.71	79.76	75.14
RGMem	**80.15**	**69.16**	67.71	**81.27**	**78.92**
Full-Context	*83.01*	*66.79*	*49.53*	*58.22*	*71.41*
gpt-4.1-mini					
RAG	37.94	37.69	48.96	61.83	51.62
LangMem	74.47	61.06	67.71	86.92	78.05
Mem0	62.41	57.32	44.79	66.47	62.47
Zep	79.43	69.16	**73.96**	83.33	79.09
RGMem	**89.58**	**78.03**	72.86	**88.91**	**86.17**
Full-Context	*88.53*	*77.70*	*71.88*	*92.70*	*87.52*

Resolving the ***stability-plasticity*** dilemma via RG-inspired multi-scale memory evolution.

5. Additional Analyses

Method	Memory Recall: Recall Facts	Memory Recall: Suggest Ideas	Memory Recall: Revisit Reasons	Reasoning & Adaptation: Track Evol.	Reasoning & Adaptation: Aligned Rec.	Reasoning & Adaptation: General. Scen.	Temporal: Latest Pref.	Summary: Overall	Summary: Avg. Tokens
Backbone: GPT-4o-mini									
w/o L0	34.35	**28.31**	72.23	47.82	70.12	43.79	62.31	47.22	**2,355**
w/o L1	70.22	15.26	77.79	50.17	59.85	41.22	60.75	54.31	4,267
w/o RG	72.35	24.27	81.82	56.72	63.54	47.15	67.88	57.37	8,479
RGMem	**77.06**	26.47	**85.29**	**67.82**	**73.66**	**56.62**	**75.47**	**63.87**	7,105

Method	Question Type: S-Hop	M-Hop	Open	Temp	Summary: Overall	Avg.Tok.
gpt-4.1-mini						
w/o L0	58.58	41.48	52.51	65.26	56.29	**1,885**
w/o L1	85.90	65.59	64.60	81.87	79.88	2,068
w/o RG	87.21	69.23	63.05	84.22	82.17	4,354
RGMem	**89.58**	**78.03**	**72.86**	**88.91**	**86.17**	3,788

Method	Recall Facts	Track Pref. Evol.	Latest Pref.
L0+GraphRAG	79.86	51.49	49.72
RGMem	**81.65**	**68.77**	**74.11**

Aggregation Mechanism	Latest Pref. (%)
w/o Δ extraction (Single Summary)	52.58
with (Σ, Δ) extraction	**74.11**

